

CAN/LIN Raw Bus Capture Measurement Plugin for Instrument Studio in LabVIEW

This plugin is an easy-to-use CAN/LIN Bus capture with sorting and filtering option. It uses the Measurement Plugin SDK.

Contents

1. The service logic main file is the **CAN-LIN Raw Bus Capture\Measurement Logic.vi**.
2. The UI part is implemented in the **CAN-LIN Raw Bus Capture UI\Measurement UI.vi** file.

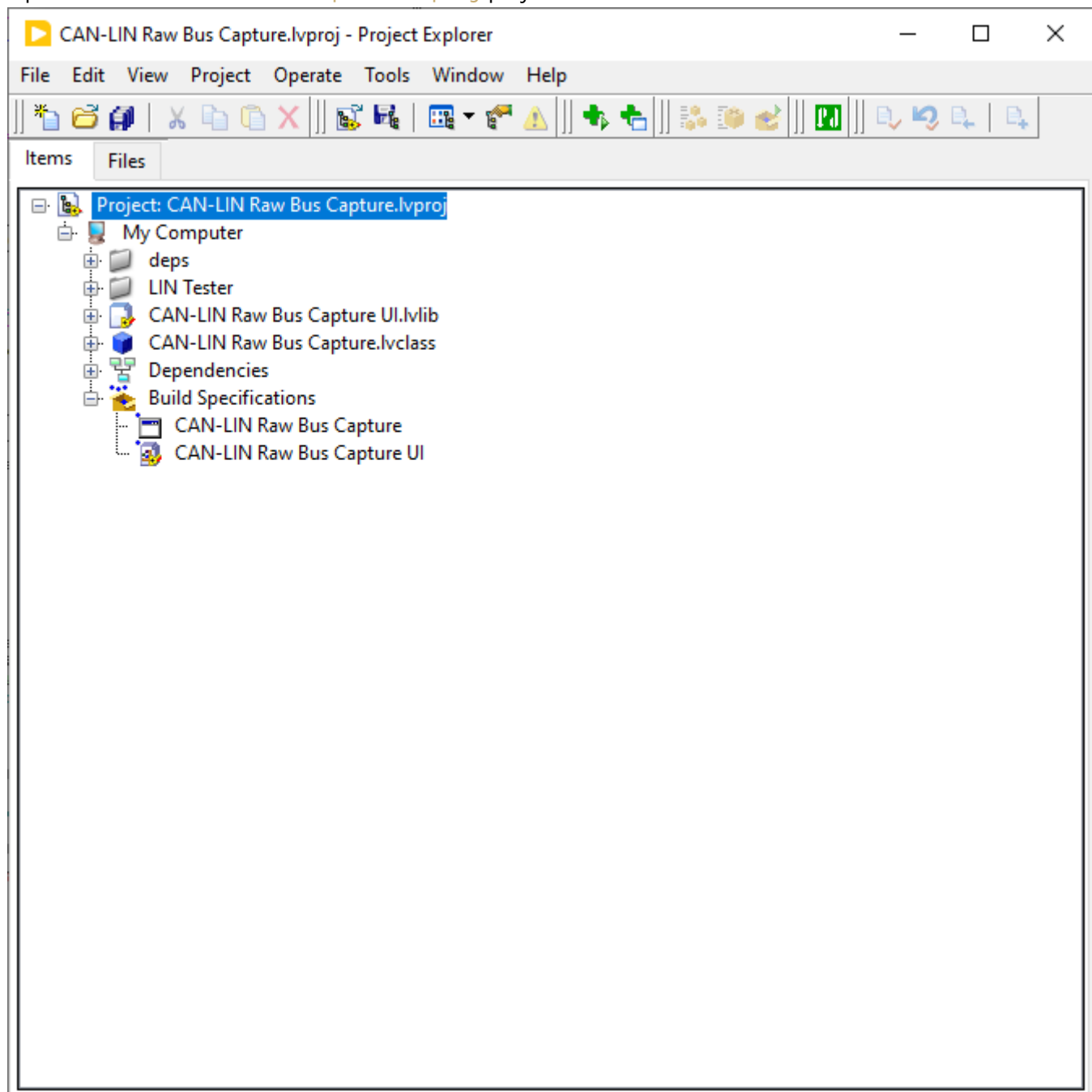
The plugin uses the **Tables and Filtering** library for sorting and filtering. It supports the OOB (Out-Of-Band) gRPC Service for interactive UI parts but right now the plugin does not use it. Keep it for future improvement.

Prerequisites

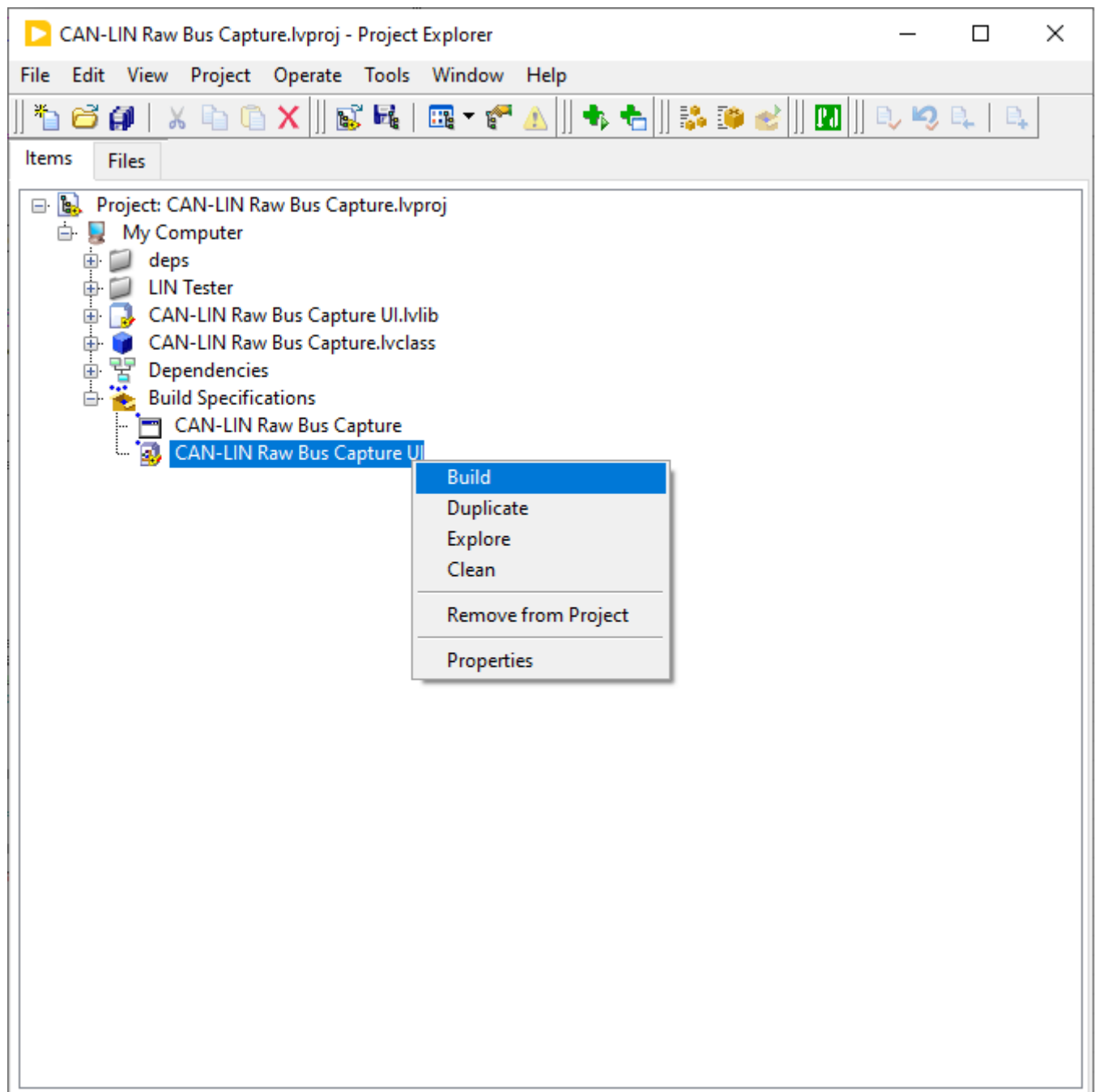
- **LabVIEW 2025 Q3**
- **Measurement Plugin SDK 3.5.0.2**
- **Instrument Studio 2025 Q4.**
- **NI PXIe-8510** with NI-XNET-CAN and/or NI-XNET-LIN transceiver

How to run this plugin

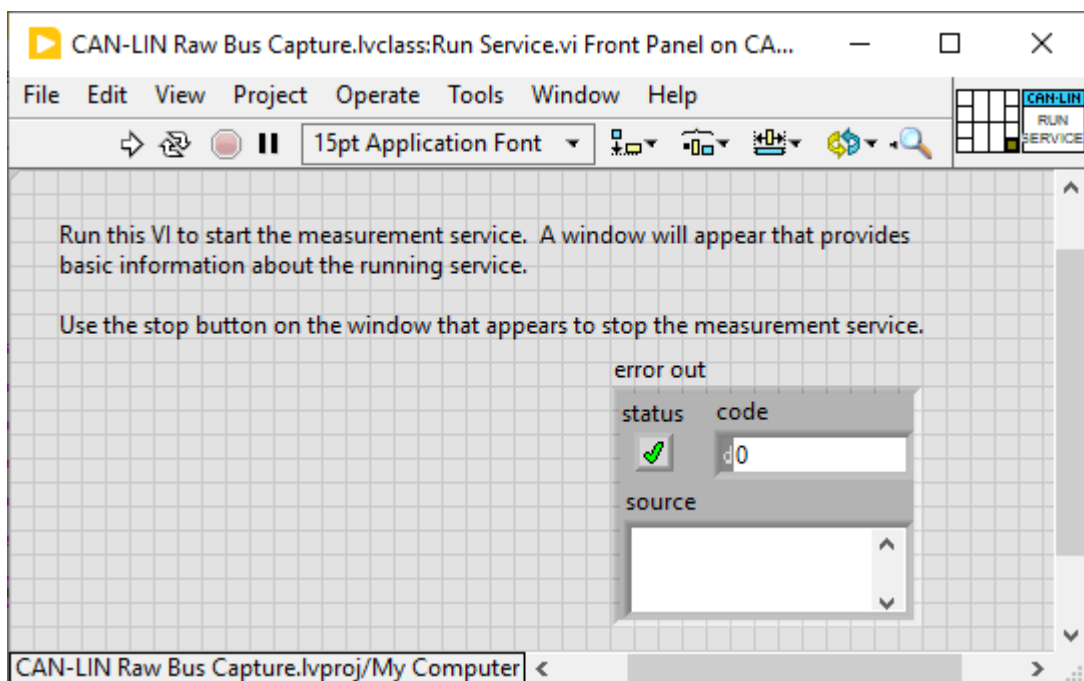
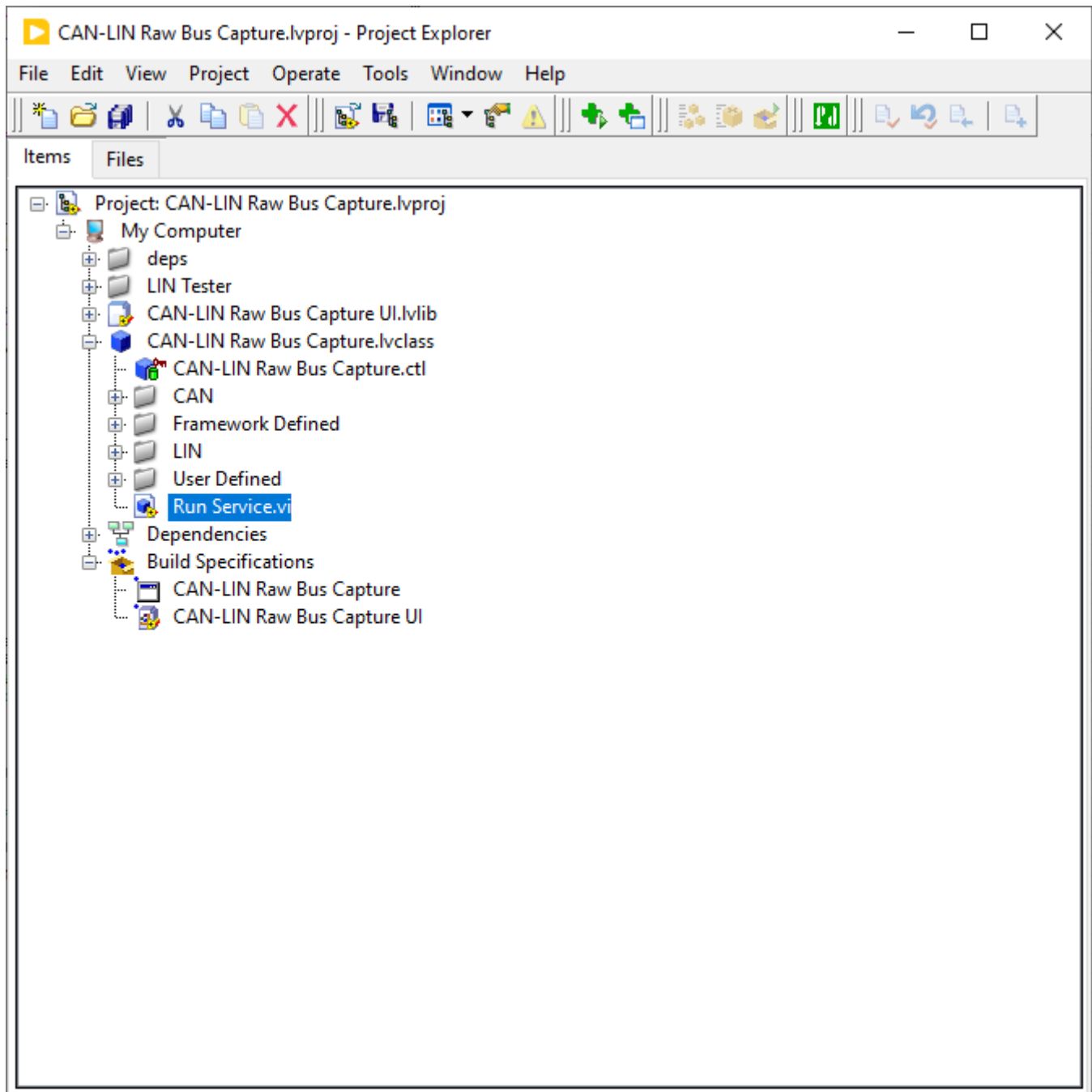
Open the **CAN-LIN Raw Bus Capture.lvproj** project in **LabVIEW 2025**.



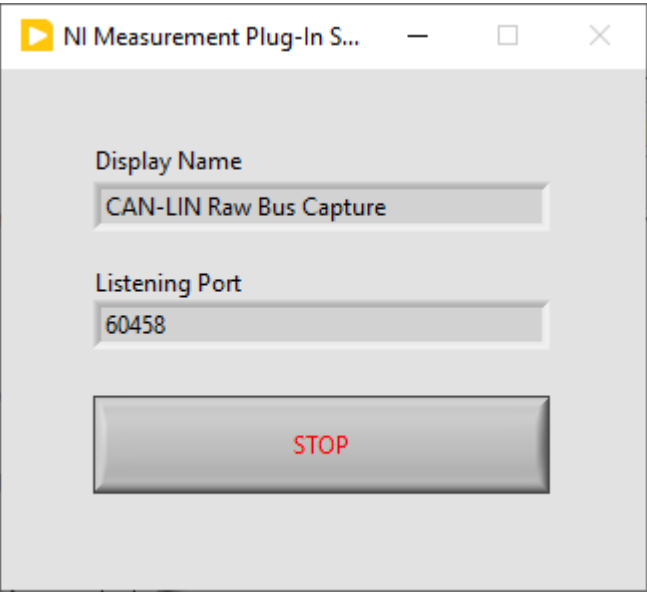
Build the **CAN-LIN Raw Bus Capture UI**.



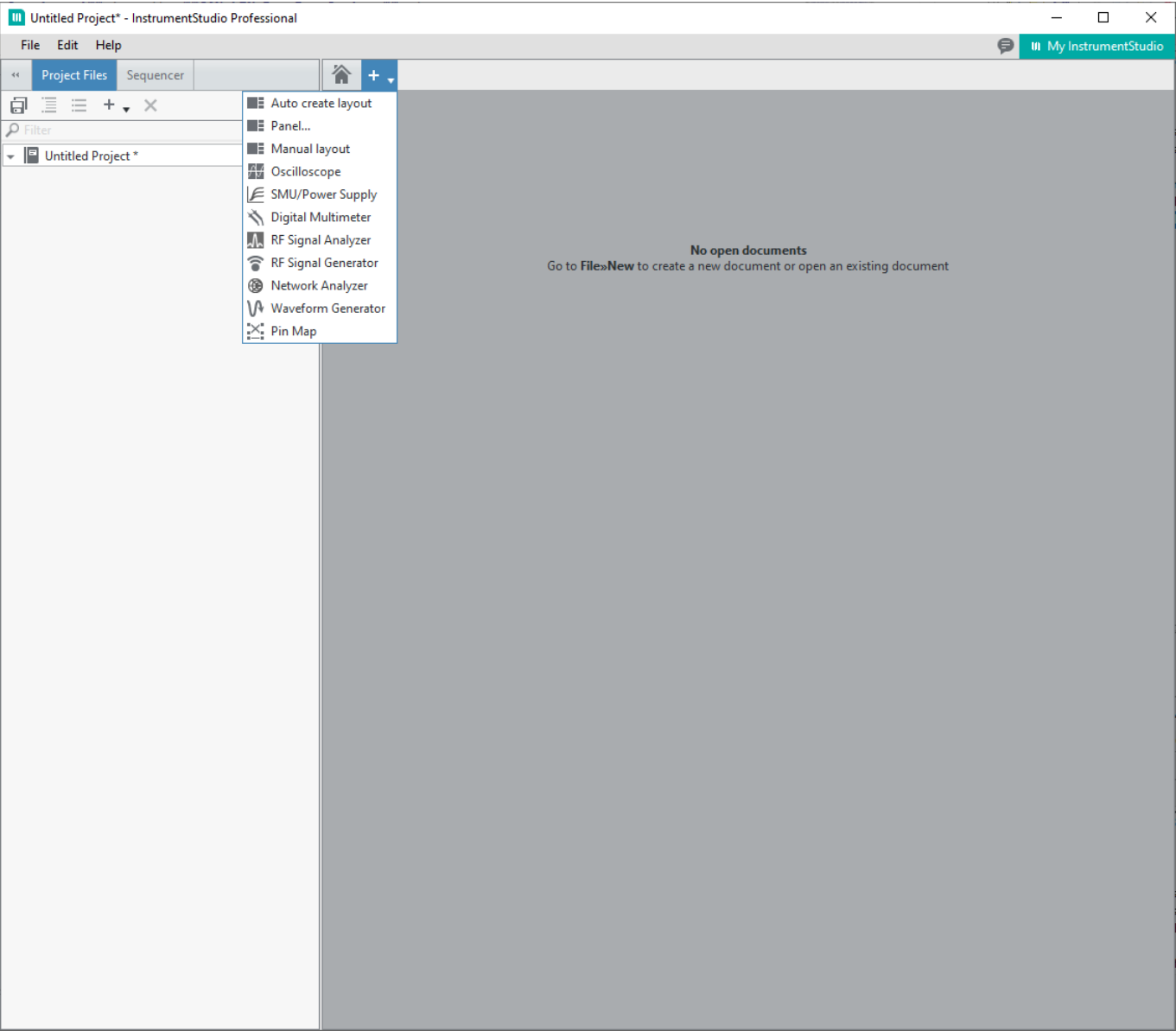
Open **Run Service.vi** from the **CAN-LIN Raw Bus Capture** class



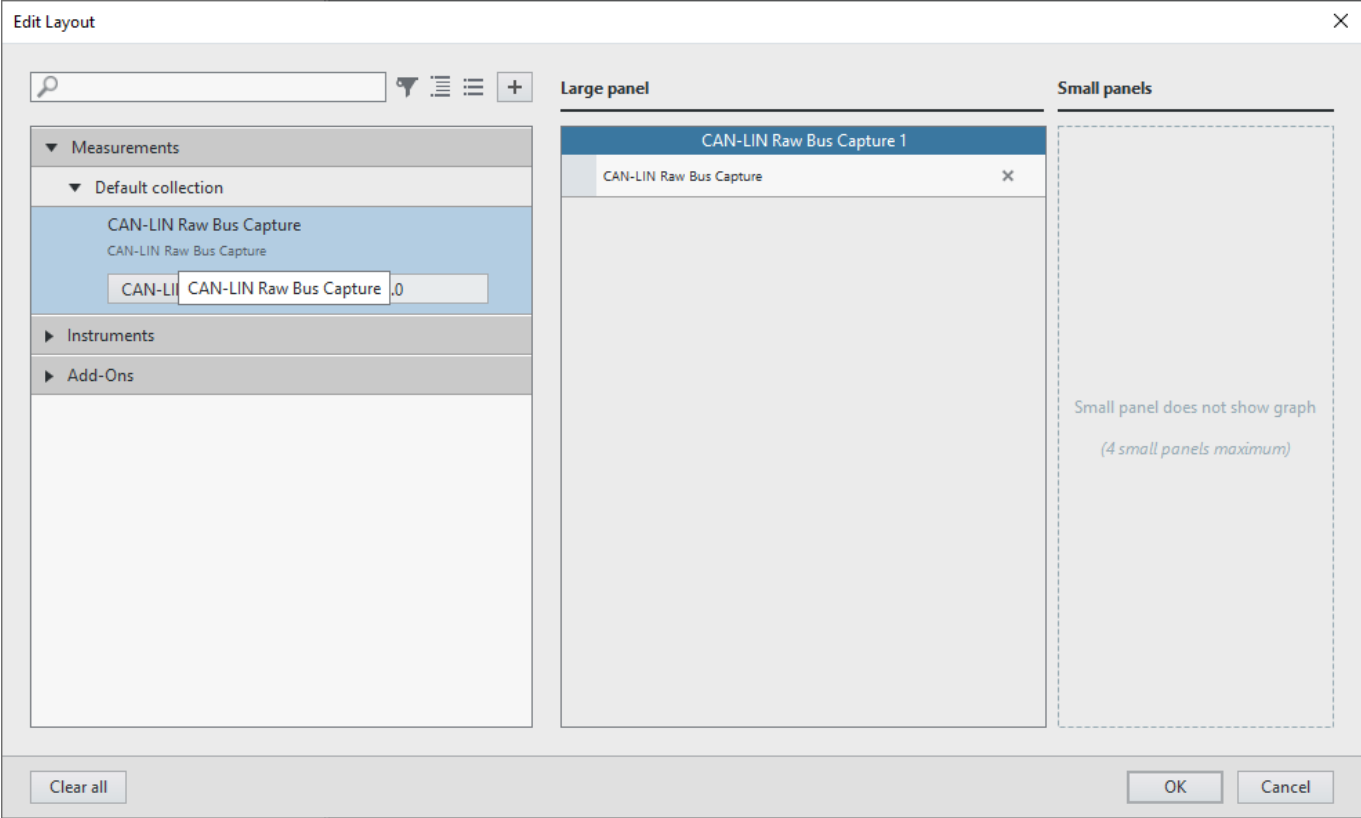
Run the VI



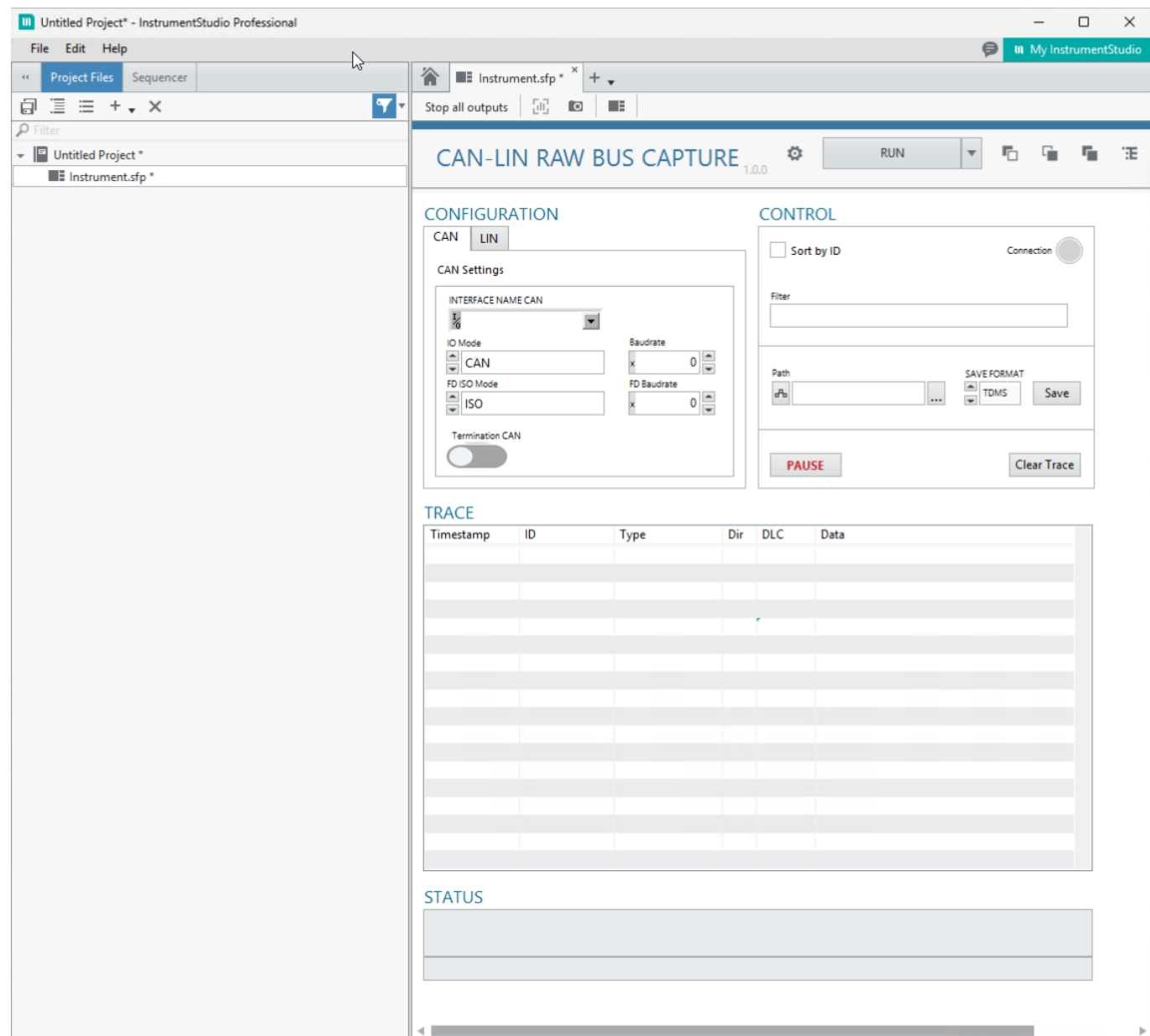
Open **Instrument Studio 2025** and create **Manual Layout**



Select **CAN-LIN Raw Bus Capture** as a large Panel



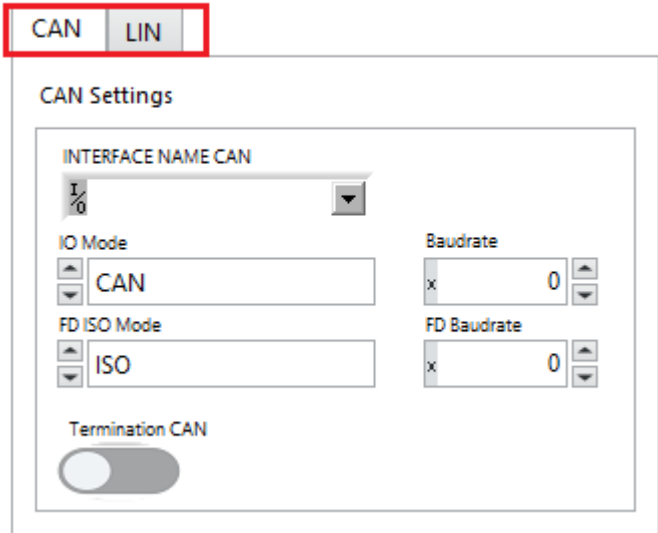
The plugin UI will shown in the Instrument Studio like this:



How to configure the plugin

Select the bus type

CONFIGURATION



CAN configuration

CONFIGURATION

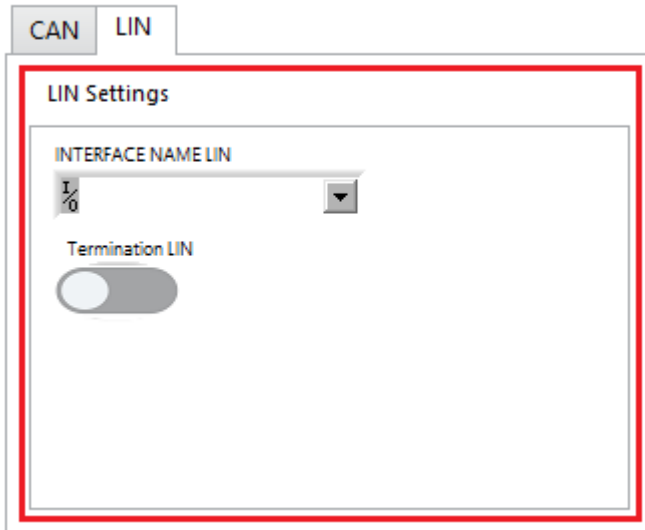
The screenshot shows a configuration window with two tabs: 'CAN' and 'LIN'. The 'CAN' tab is active. The 'CAN Settings' section contains the following controls:

- INTERFACE NAME CAN:** A dropdown menu showing 'I/O'.
- IO Mode:** A dropdown menu showing 'CAN'.
- FD ISO Mode:** A dropdown menu showing 'ISO'.
- Baudrate:** A numeric input field showing '0'.
- FD Baudrate:** A numeric input field showing '0'.
- Termination CAN:** A toggle switch currently set to 'off'.

1. Select the CAN interface
2. Select the IO Mode:
 - CAN (Standard Controller Area Network)
 - CAN FD (Flexible Data-Rate)
 - CAN FD+BRS (Flexible Data-Rate and Bit Rate Switch)
3. Set baudrate (hexadecimal value)
 - 1E848 for 125 kbps
4. Set FD ISO Mode (optional when IO Mode is set to CAN FD or CAN FD+BRS)
 - ISO
 - Non ISO
 - Legacy ISO
5. Set FD baudrate (hexadecimal value) (optional when IO Mode is set to CAN FD or CAN FD+BRS)
 - F4240 for 1 mbps
6. Turn on the termination resistor if it is needed for the bus.

LIN configuration

CONFIGURATION

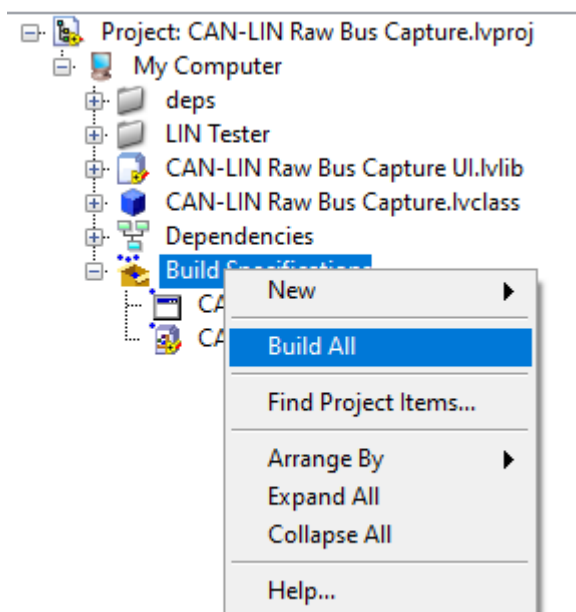


1. Select the LIN(Local Interconnect Network) interface
2. Turn on the termination resistor if it is needed for the bus.

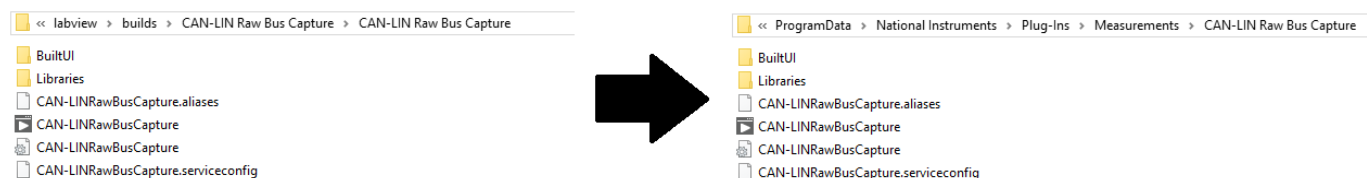
*Note: The plugin connects to the LIN bus as a slave node with 128 kpbs baudrate.

How to deploy the plugin (local only)

Build the **CAN-LIN Raw Bus Capture** and the **CAN-LIN Raw Bus Capture UI**

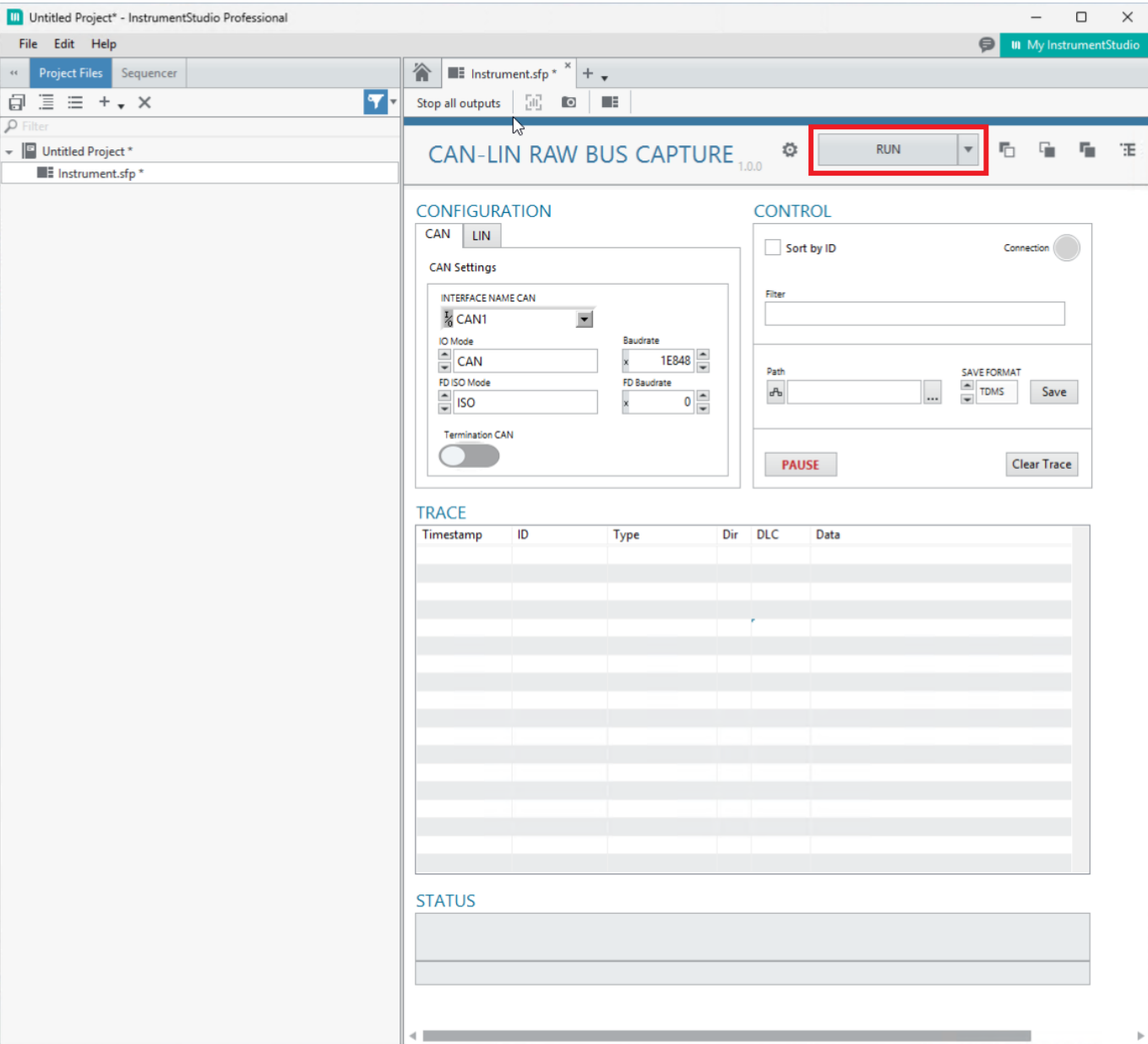


Copy the `src\labview\builds\CAN-LIN Raw Bus Capture\CAN-LIN Raw Bus Capture` folder under the `C:\ProgramData\National Instruments\Plug-Ins\Measurements` location.



How to start and use the plugin

Start the plugin with the **RUN** button in the Instrument Studio.



It connects to the selected bus and starts listening. The service updates the UI every 150ms for CAN, and 100ms for LIN. The connection LED turns on. It shows the captured messages in the trace table and the bus status in the status area at the bottom.

CAN-LIN RAW BUS CAPTURE1.0.0

⚙️

STOP

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⚙️

CANLIN

CAN Settings

INTERFACE NAME CAN

1/10

CAN1

▼

IO Mode

⬆️

CAN

⬆️

FD ISO Mode

⬆️

ISO

⬆️

Baudrate

x

1E848

⬆️

FD Baudrate

x

F4240

⬆️

Termination CAN

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☐ Sort by ID

Connection

🟢

Filter

Path

📁

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SAVE FORMAT

⬆️

TDMS

⬆️

Save

PAUSE

Clear Trace

TRACE

Timestamp	ID	Type	Dir	DLC	Data
2026-01-15 10:28:24.302	0x7FF	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:28:35.426	0xCEFFA	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:28:48.156	0x300	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:28:53.457	0x7	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:29:04.685	0x22F	CAN Data	Rx	8	07 07 07 04 05 06 07 08
2026-01-15 10:29:18.236	0xA00E	CAN Data	Rx	4	07 07 07 04
2026-01-15 10:29:33.798	0xB00E	CAN Data	Rx	4	02 02 07 04
2026-01-15 10:29:52.684	0x101	CAN Data	Rx	4	01 02 03 04
2026-01-15 10:30:00.055	0x10	CAN Remote	Rx	4	00 00 00 00

STATUS

Communication state: Error Active

Last error: None, Receive error counter: 0

No Fault, Sleep inactive

The user can interact with the plugin by the control area such as sorting, filtering, pause / resume and clear the trace table or save log files in TDMS or CSV format.

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CONTROL

☐ Sort by ID

Connection 

Filter

Path



SAVE FORMAT

  TDMS

Save

PAUSE

Clear Trace

Sorting

The default order of the messages is by timestamp, it can change by the **Sort by ID** checkbox.

CAN-LIN RAW BUS CAPTURE1.0.0

STOP

CANLIN

CAN Settings

INTERFACE NAME CAN

1/10

CAN1

IO Mode

CAN

FD ISO Mode

ISO

Baudrate

x

1E848

FD Baudrate

x

F4240

Termination CAN

☒ Sort by ID

Connection

Filter

Path

...

SAVE FORMAT

TDMS

Save

PAUSE

Clear Trace

TRACE

Timestamp	ID	Type	Dir	DLC	Data
2026-01-15 10:28:53.457	0x7	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:30:00.055	0x10	CAN Remote	Rx	4	00 00 00 00
2026-01-15 10:29:52.684	0x101	CAN Data	Rx	4	01 02 03 04
2026-01-15 10:29:04.685	0x22F	CAN Data	Rx	8	07 07 07 04 05 06 07 08
2026-01-15 10:28:48.156	0x300	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:28:24.302	0x7FF	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:29:18.236	0xA00E	CAN Data	Rx	4	07 07 07 04
2026-01-15 10:29:33.798	0xB00E	CAN Data	Rx	4	02 02 07 04
2026-01-15 10:28:35.426	0xCEFFA	CAN Data	Rx	8	01 07 07 04 05 06 07 08

STATUS

Communication state: Error Active

Last error: None, Receive error counter: 0

No Fault, Sleep inactive

Filter

Use the headers of the trace table to set conditions what to filter.

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CAN-LIN RAW BUS CAPTURE1.0.0

⚙️

STOP

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📄

CANLIN

CAN Settings

INTERFACE NAME CAN

1/0

CAN1

▼

IO Mode

⬆️

CAN

⬆️

Baudrate

x

1E848

⬆️

FD ISO Mode

⬆️

ISO

⬆️

FD Baudrate

x

F4240

⬆️

Termination CAN

⬅️

☒ Sort by ID

Connection

🟢

Filter

DLC == 8 AND ID < 0x300

Path

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SAVE FORMAT

⬆️

TDMS

⬆️

Save

PAUSE

Clear Trace

TRACE

Timestamp	ID	Type	Dir	DLC	Data
2026-01-15 10:28:53.457	0x7	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:29:04.685	0x22F	CAN Data	Rx	8	07 07 07 04 05 06 07 08

CAN-LIN RAW BUS CAPTURE1.0.0

⚙️

STOP

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📄

📄

CANLIN

CAN Settings

INTERFACE NAME CAN

1/0

CAN1

▼

IO Mode

⬆️

CAN

⬆️

Baudrate

x

1E848

⬆️

FD ISO Mode

⬆️

ISO

⬆️

FD Baudrate

x

F4240

⬆️

Termination CAN

⬅️

☒ Sort by ID

Connection

🟢

Filter

Data == "01 07 07 04 05 06 07 08"

Path

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...

SAVE FORMAT

⬆️

TDMS

⬆️

Save

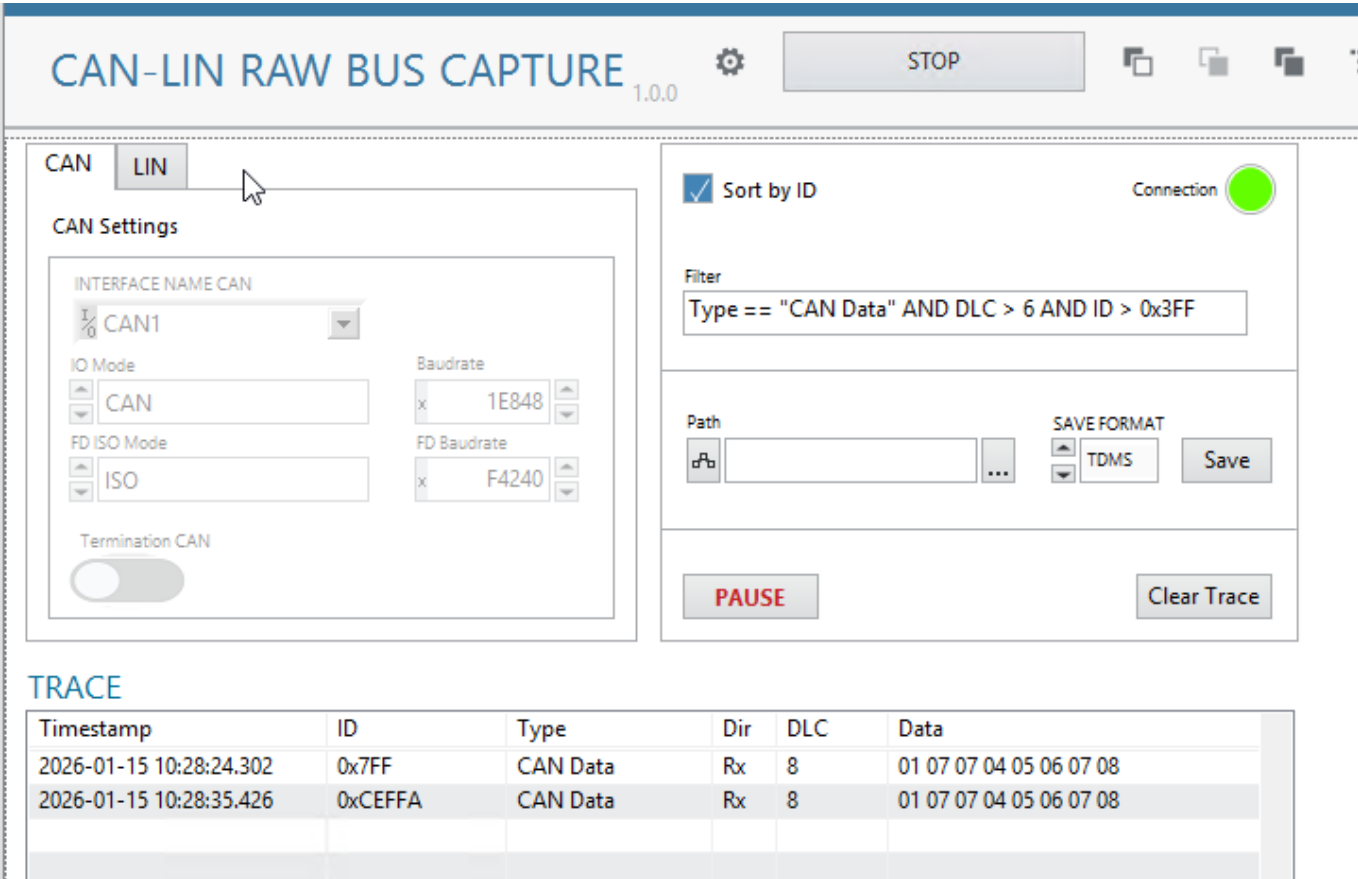
PAUSE

Clear Trace

TRACE

Timestamp	ID	Type	Dir	DLC	Data
2026-01-15 10:28:53.457	0x7	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:28:48.156	0x300	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:28:24.302	0x7FF	CAN Data	Rx	8	01 07 07 04 05 06 07 08
2026-01-15 10:28:35.426	0xCEFFA	CAN Data	Rx	8	01 07 07 04 05 06 07 08

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Save trace to log file (TDMS, CSV)

The user can set a path in the control area and save the trace data to CSV or TDMS file with the **SAVE** button. The file name auto generated as {BusType_tracelog_YYYYMMDD_HHIISSUUUU} User got response message at the bottom of the Status area.

Pause / Resume capture

With this button the user can pause (suspend) the trace table update. At the background the plugin continue capturing the messages and update the table on resume.

Details for Developers

Measurement Logic.vi

This VI starts when the users clicks on the RUN button in the Instrument Studio. It gets the Measurement Configuration.ctl data from the UI and starts with the initialization of the selected XNET interface. For LIN the baudrate is fixed to 19200 bps. Initialization / Open Session for CAN is implemented in the "Open and Setup CAN Session" for LIN is in the "Open and Setup LIN Session" subVI.

LIN

The VI starts a loop with 100ms rate and it reads messages (frames) and the state from the bus in each cycle and sends back to the UI (propagate) with the "Update Results.vi". It converts the messages to strings with the "Convert LIN frames to Trace log.vi" subVI.

CAN

The VI starts a loop with 1ms rate and it reads messages (frames) from the bus and stores them in a buffer. It sends back (propagate) the collected messages with the bus state 150ms rate by the "Update Results.vi". It converts the messages to strings with the "Convert CAN frames to Trace log.vi" subVI.

Measurement Results.ctl

It contains the data what the logic sends back periodically to the UI. It contains:

- Message buffer (2D array of strings): the collected messages from the bus since the last "Update Result.vi" call
- Status (string) : actual bus state.
- Connection (boolean): true while the logic runs.
- OOB gRPC port (int): the port number where the OOB gRPC is listening, the UI should connect to it and propagate data to the logic. This plugin is not used OOB functionality, it was implemented for future use.

Measurement UI.vi

The block diagram has two loop. The bottom one manage the OOB connection and data exchange from UI to Logic. The plugin does not use it in this version. The main loop on the top holds the captured messages and the buffered messages on pause, and manages the following:

- Events
- Initialize: runs on start
- Post UI Update: runs each time when UI got data from logic.
- Exit: runs on exit
- Handle Error

The "Captured Data" array contains each messages which was received from the logic in normal run (not on pause). The "Buffer for pause" array contains the messages from the logic under the plugin is paused. On resume the plugin copies the "Buffer for pause" data to the "Captured Data" array.

Post UI Update

It checks the received message array (Message Buffer) size and

- (in normal, not paused run) when it contains data updates the "Captured Data" array with them checks the new size of the "Captured Data" array, keep latest 10.000 element, than filter and order data and updates the Trace table.
- (in pause) when it contains data append it to the "Buffer for pause" array

Events

- Interface Name CAN: Set string value for "CAN interface" when user changes CAN interface.
- Interface Name LIN: Set string value for "LIN interface" when user changes LIN interface.
- Settings: Set int value for "Bus Type" when user changes Configuration tab.
- Clear Trace: Remove data from Trace table and from "Captured Data" array. It scrolls up the Trace table too.
- SortByID, Filter change: Updates the Trace table based on the filter and the SortByID value.

- PauseResume:
 - When user clicks on **PAUSE**, it clears the "Buffer for pause" array.
 - When user clicks on **RESUME**, it append the "Buffer for pause" array to the "Captured Data" array, check the new size and keep the latest 10.000 element, than filter and order the data and updates the Trace table. At the end it removes each element from the "Buffer for pause" array.
- Save: based on the SAVE FORMAT value it saves the Trace table to CSV or TDMS file when the path is valid. The log file name is auto generated. For TDMS file it converts the string values from the Trace Table to raw frame with "Convert CAN Trace to Raw Frame.vi" and "Convert LIN Trace to Raw Frame.vi".